BA08 PROJECT

Price prediction of Airbnb in Bordeaux

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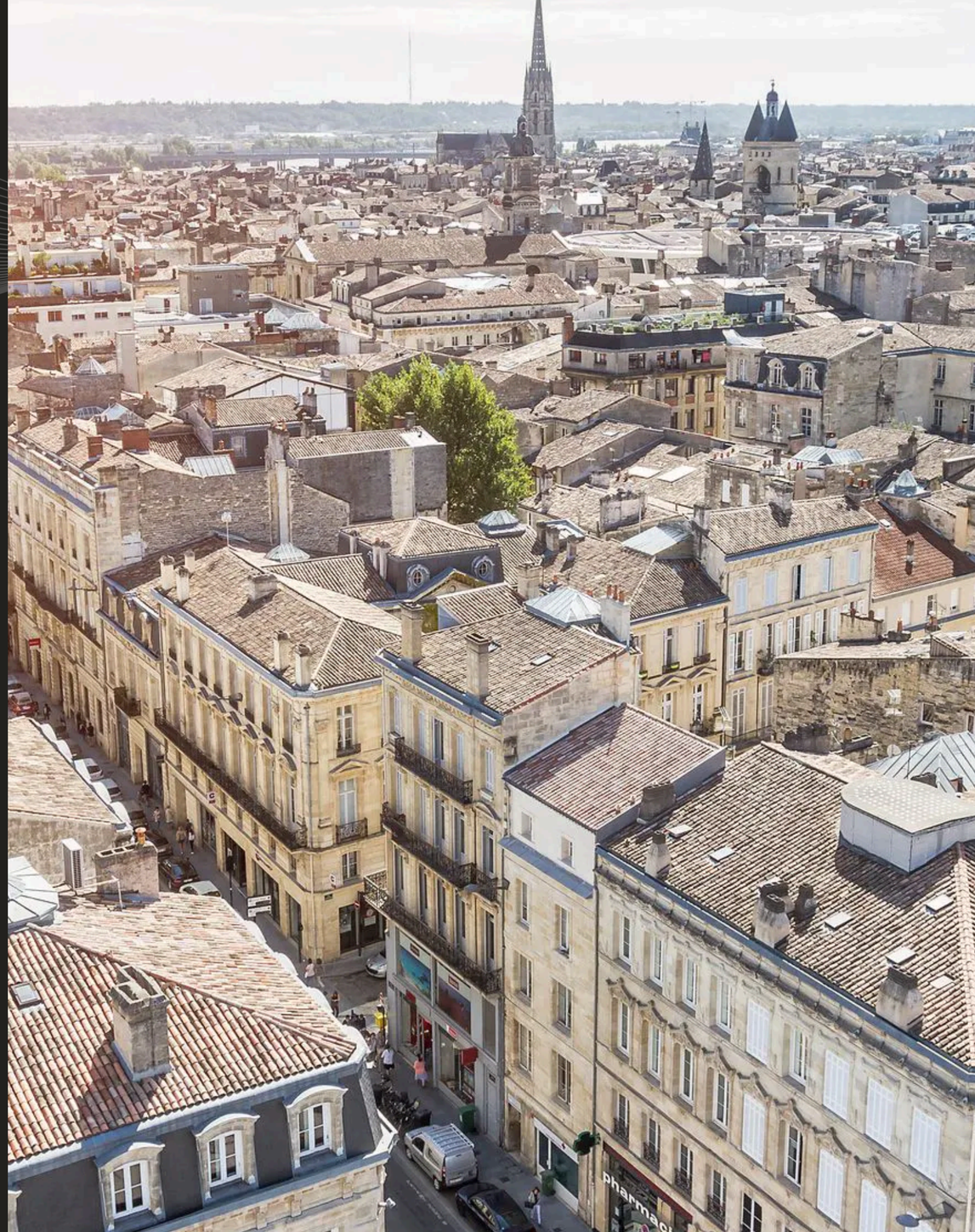

Introduction

Project Objective:

Prediction price = f (X1,X2,X3……)

Train a model that can predict the rental price of an appartement/studio in Bordeaux

Data source: Airbnb website -> file.csv

The file.csv is available on the website:

https://insideairbnb.com/get-the-data/

Model Applications:

1. Investment Analysis
2. Price Optimization for Hosts

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- 1.1 Data Transformation
- 1.2 Data Cleaning

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- 2.2 The Relationship between Price and Other Data

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1.1 Data Transformation

	B
49	Condo in Mérignac · ★5.0 · 1 bedroom · 2 beds · 1 bath
50	Rental unit in Bordeaux · ★New · 2 bedrooms · 3 beds · 1 bath
51	Home in Ambarès-et-Lagrave · ★New · 3 bedrooms · 4 beds · 2 baths
52	Home in Bordeaux · ★New · 1 bedroom · 2 beds · 1 bath
53	Rental unit in Bordeaux · 1 bedroom · 2 beds · 1 bath
54	Rental unit in Bordeaux · ★4.50 · 1 bedroom · 3 beds · 1 bath
55	Rental unit in Bordeaux · ★5.0 · 2 bedrooms · 3 beds · 1 bath
56	Rental unit in Le Bouscat · 1 bedroom · 1 bed · 1 bath
57	Rental unit in Le Bouscat · ★New · 1 bedroom · 1 bed · 1 bath
58	Rental unit in Talence · ★New · 1 bedroom · 1 bed · 1.5 baths
59	Rental unit in Cenon · 1 bedroom · 1 bed · 1 bath
60	Rental unit in Bordeaux · 1 bedroom · 1 bed · 1 bath
61	Rental unit in Bordeaux · ★5.0 · 1 bedroom · 3 beds · 1.5 baths
62	Rental unit in Lormont · ★4.89 · 1 bedroom · 1 bed · 1 bath
63	Rental unit in Bordeaux · ★4.54 · 2 bedrooms · 2 beds · 1 bath
64	Rental unit in Bordeaux · ★4.67 · 1 bedroom · 2 beds · 1 bath
65	Rental unit in Le Bouscat · ★New · 2 bedrooms · 2 beds · 1 bath
66	Rental unit in Bordeaux · ★4.67 · 1 bedroom · 1 bed · 1 bath
67	Home in Bordeaux · 3 bedrooms · 3 beds · 3.5 baths
68	Rental unit in Bordeaux · ★4.29 · 2 bedrooms · 2 beds · 1 bath
69	Rental unit in Bordeaux · 1 bedroom · 1 bed · 1 bath
70	Rental unit in Talence · ★4.67 · 1 bedroom · 2 beds · 1 bath
71	Rental unit in Talence · ★4.67 · Studio · 1 bed · 1 bath
72	Rental unit in Talence · ★5.0 · 1 bedroom · 2 beds · 1 bath
73	Rental unit in Bordeaux · 3 bedrooms · 3 beds · 2 baths
74	Home in Ambarès-et-Lagrave · ★4.92 · 3 bedrooms · 3 beds · 1 bath

	note	number_of_bedrooms	number_of_beds	number_of_baths	type_of_baths
49	-1.00	3	4	2	private
50	-1.00	1	2	1	private
51	NaN	1	2	1	private
52	4.50	1	3	1	private
53	5.00	2	3	1	private
54	NaN	1	1	1	private
55	-1.00	1	1	1	private
56	-1.00	1	1	1.5	private
57	NaN	1	1	1	private
58	NaN	1	1	1	private
59	5.00	1	3	1.5	private
60	4.89	1	1	1	private
61	4.54	2	2	1	private
62	4.67	1	2	1	private
63	-1.00	2	2	1	private
64	4.67	1	1	1	private
65	NaN	3	3	3.5	private
66	4.29	2	2	1	private
67	NaN	1	1	1	private
68	4.67	1	2	1	private
69	4.67	1	1	1	private
70	5.00	1	2	1	private
71	NaN	3	3	2	private
72	4.92	3	3	1	private
73	NaN	1	1	1	private

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1.2 Data cleaning

Original data

id	name	host_id	host_name	neighbourhood_group	neighbourhood
1.05E+18	Rental unit in Bordeaux · ★New · 2 bedrooms · 1 bath	550766188	Marie	Bordeaux	Centre ville (Bordeaux)
1.04E+18	Rental unit in Bordeaux · ★New · 1 bedroom · 1 bed · 1 bath	550478445	Philippe	Bordeaux	Saint Augustin - Tauzin - Alphonse I
1.04E+18	Rental unit in Bordeaux · ★New · 1 bedroom · 1 bed · 1 bath	550336355	Thibaut	Bordeaux	Chartrons - Grand Parc - Jardin Pu
1.04E+18	Home in Talence · ★New · Studio · 1 bed · 1 bath	549947476	Amélie	Talence	Talence
1.04E+18	Home in Floirac · ★New · 2 bedrooms · 2 beds · 1 bath	549901957	Caroline	Floirac	Floirac
1.04E+18	Rental unit in Bordeaux · ★New · 2 bedrooms · 2 beds · 1 bath	549875028	Diego	Bordeaux	Bordeaux Sud
1.04E+18	Rental unit in Talence · ★New · Studio · 1 bed · 1 bath	549874399	Frederic	Talence	Talence
1.04E+18	Rental unit in Parempuyre · ★New · 1 bedroom · 1 bed · 1 bath	549597628	Francois	Parempuyre	Parempuyre
1.04E+18	Rental unit in Pessac · ★New · 4 bedrooms · 4 beds · 3 baths	548984748	Cedric	Pessac	Nos
1.04E+18	Rental unit in Bordeaux · ★New · Studio · 1 bed · 1 bath	548767621	Elise	Bordeaux	Centre ville (Bordeaux)
1.03E+18	Home in Le Taillan-Médoc · ★New · Studio · 1 bed · 1 bath	548124533	Solenn	Le Taillan-Mdoc	Le Taillan-Mdoc

latitude	longitude	room_type	price	minimum_nights	number_of_reviews	last_review	reviews_per_month	calculated_host_listings_count
44.83572	-0.56914	Entire home/apt	113	1	0			1
44.8334155	-0.5832504	Entire home/apt		3	0			1
44.8507023	-0.5715281	Entire home/apt	120	2	0			1
44.81587403	-0.602802026	Entire home/apt	50	1	0			1
44.8382825	-0.5369384	Entire home/apt		2	0			1
44.82916246	-0.573463705	Entire home/apt	129	1	0			1
44.8159338	-0.5786466	Entire home/apt	47	1	0			1
44.94432	-0.60872	Entire home/apt	46	2	1	2023/12/13	1	1
44.81368	-0.63599	Entire home/apt	139	1	0			1
44.83688	-0.58172	Entire home/apt	70	1	0			1
44.90914533	-0.684296647	Entire home/apt	60	1	0			1

availability_365	number_of_reviews_ltm	license
20	0	33063010974EB
139	0	330630109619D
242	0	33063009735B6
159	0	
210	0	
18	0	3.31E+12

17 columns (data label)

11853 rows (num of appartements)

1.2 Data cleaning

1. Delete the data not useful

A
id
1.05E+18
1.04E+18
1.04E+18
1.04E+18
1.04E+18
1.04E+18
1.04E+18
1.04E+18
1.04E+18
1.04E+18

D
host_name
Marie
Philippe
Thibaut
Amélie
Caroline
Diego
Frederic
Francois
Cedric

R
license
33063010974EB
330630109619D
3306300973586
3.31E+12

Number of columns: 17 - 3 = 14

1.2 Data cleaning

2. Handle the missing data

```
data.isnull().sum()
```

note	3177
number_of_bedrooms	15
number_of_beds	120
number_of_baths	1286
type_of_baths	0
host_id	0
neighbourhood_group	0
neighbourhood	0
latitude	0
longitude	0
room_type	0
price	3692
minimum_nights	0
number_of_reviews	0
reviews_per_month	1913
calculated_host_listings_count	0
availability_365	0
number_of_reviews_ltm	0

dtype: int64

-51.6% data

```
data.isnull().sum()
```

note	0
number_of_bedrooms	0
number_of_beds	0
number_of_baths	0
type_of_baths	0
host_id	0
neighbourhood_group	0
neighbourhood	0
latitude	0
longitude	0
room_type	0
price	0
minimum_nights	0
number_of_reviews	0
reviews_per_month	0
calculated_host_listings_count	0
availability_365	0
number_of_reviews_ltm	0

dtype: int64

Before: 11853 appartements/studios

After: 5736 appartements/studios

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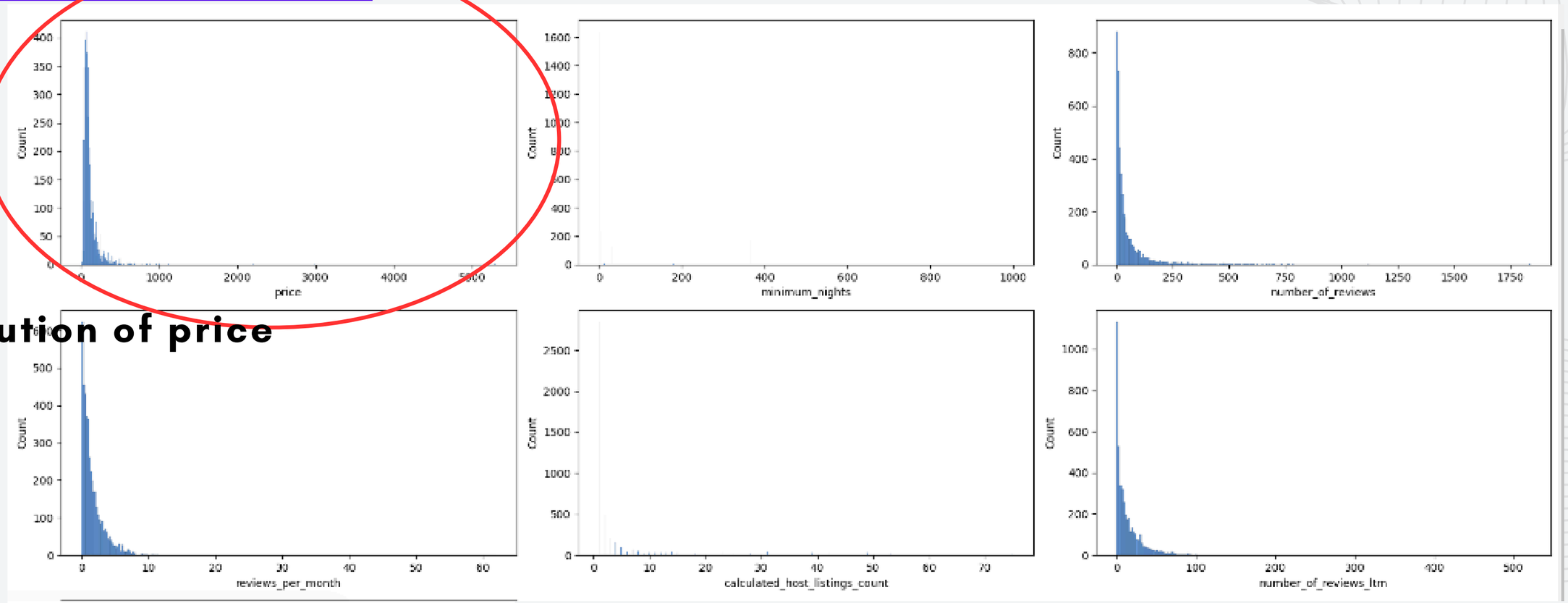
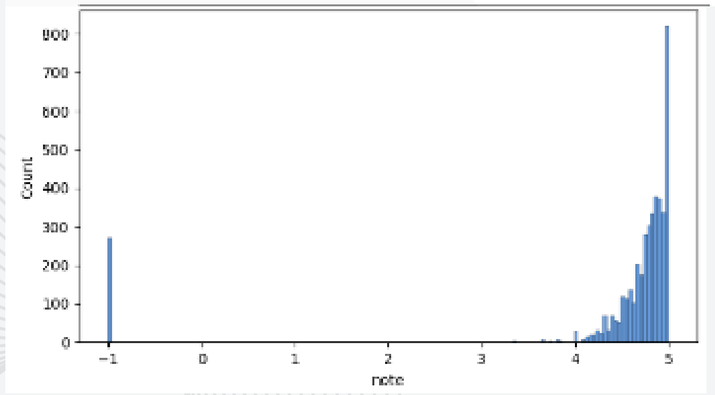
2.1 Distribution of Data

Type 1 : Long-Tail Data

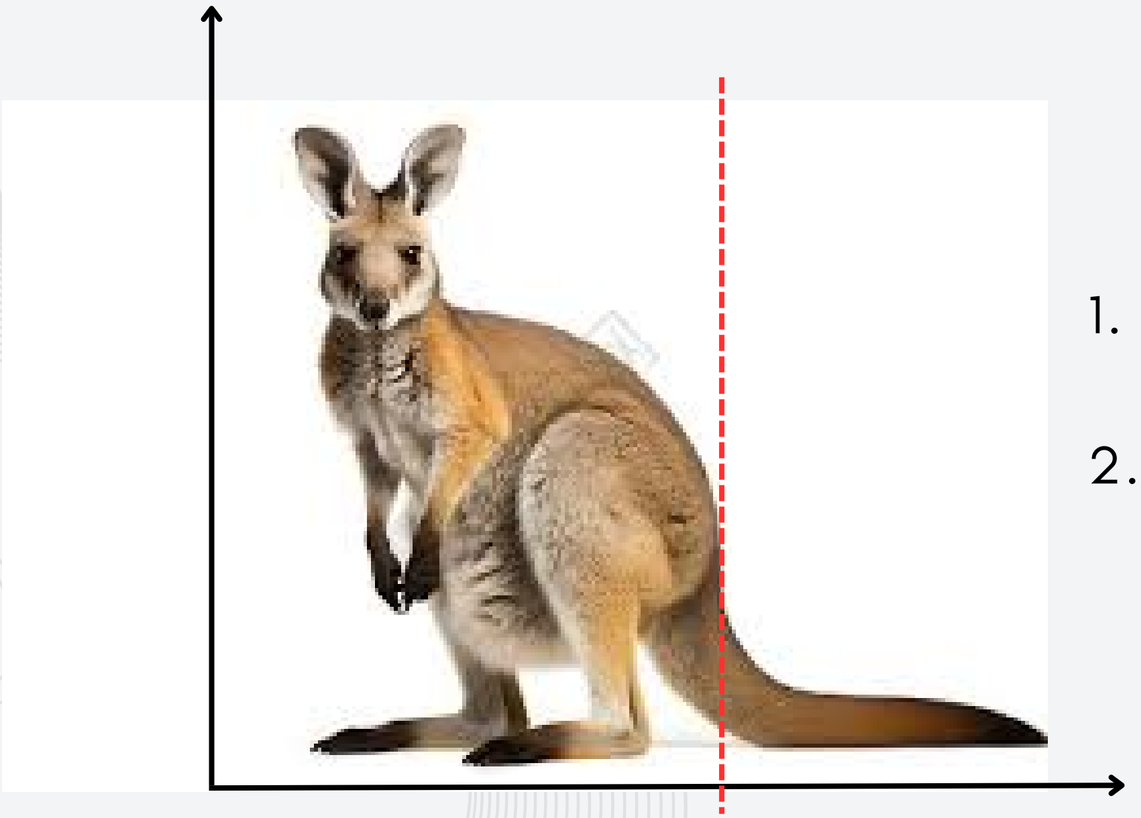


Distribution of price



2.1 Distribution of Data

Type 1 : Long-Tail Data



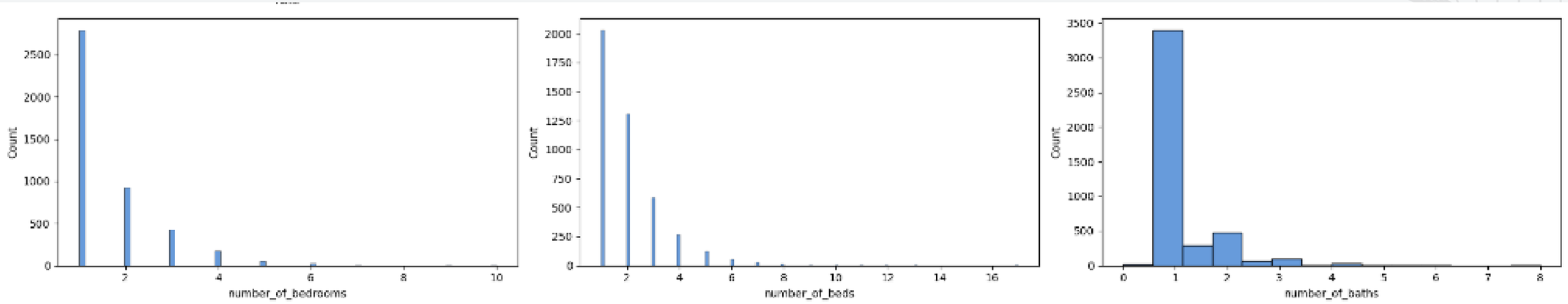
Action:

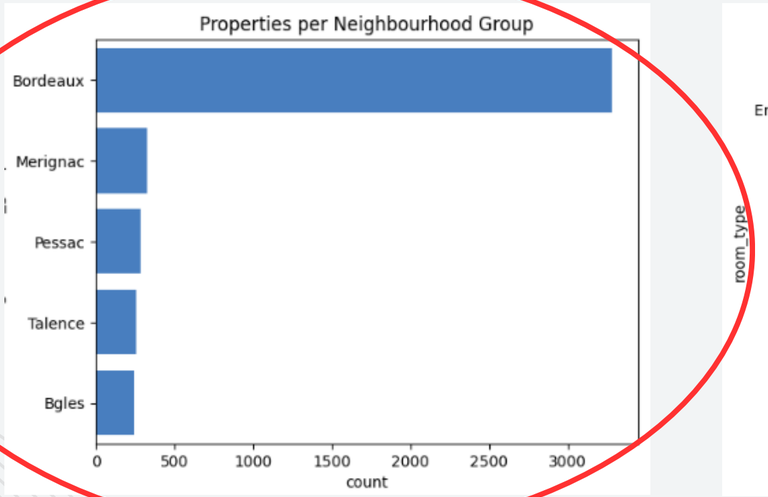
1. Price -> Low price and high price (bound: 100 euros)

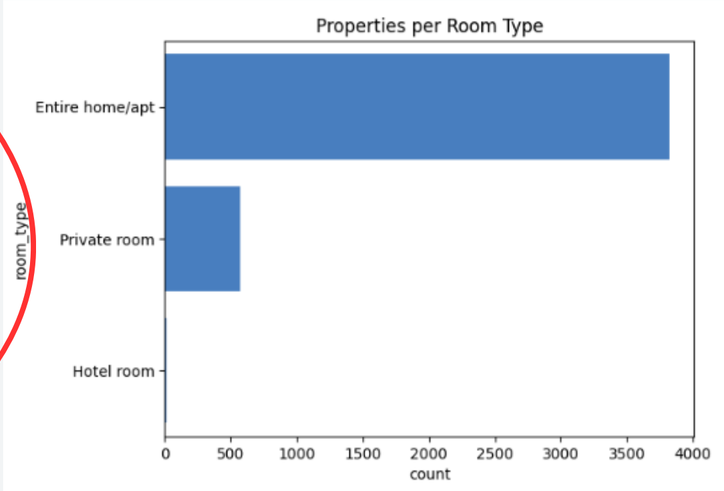
2. Log transformation: $\text{Data} = \log_{10}(\text{Data})$

2.1 Distribution of Data

Type 2 : Data with only a few categories







2.1 Distribution of Data

Type 2 : Data with only a few categories

Action:

Use **One-Hot Encoding** to convert categorical variables into numerical representations

category_A	category_B	category_C
1	0	0
0	1	0
0	0	1

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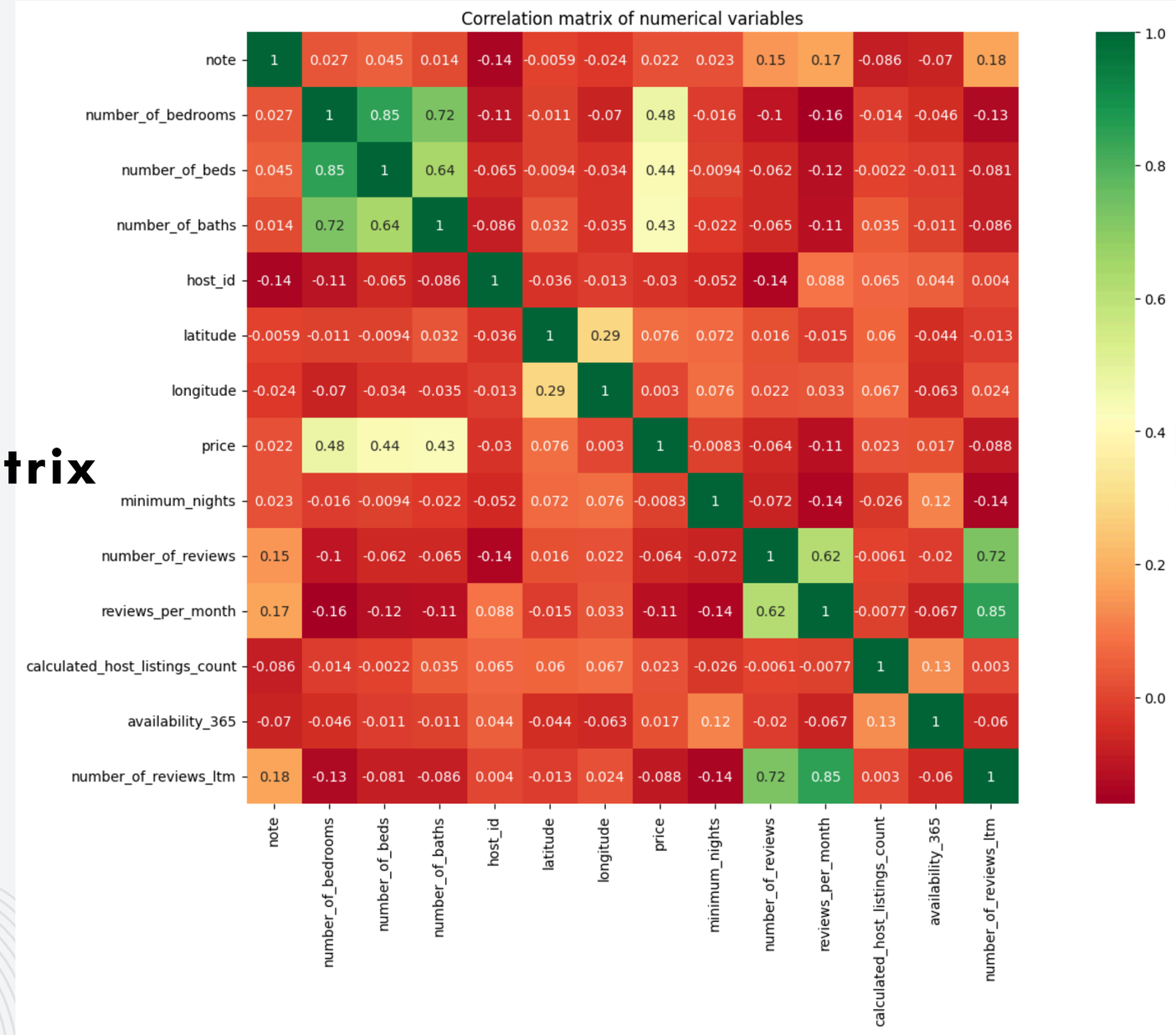
2.2 The Relationship between Price and Other Data

Correlation matrix



2.2 The Relationship between Price and Other Data

Principal of the task:

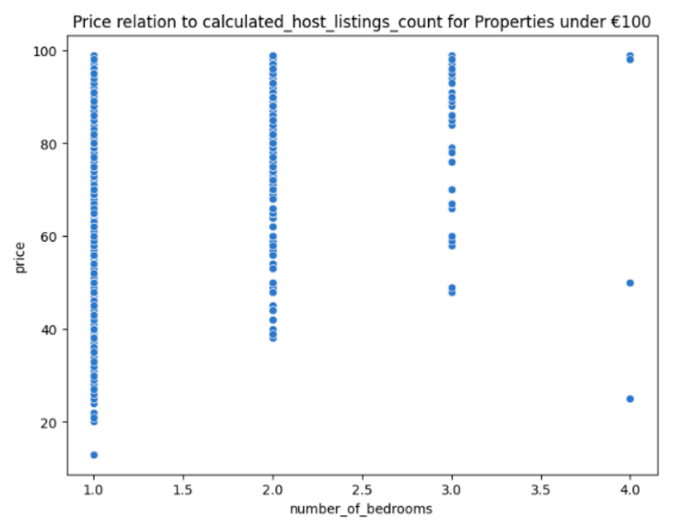
X1 : One of the labels in the Dataframe

Then: X2, X3...Xn

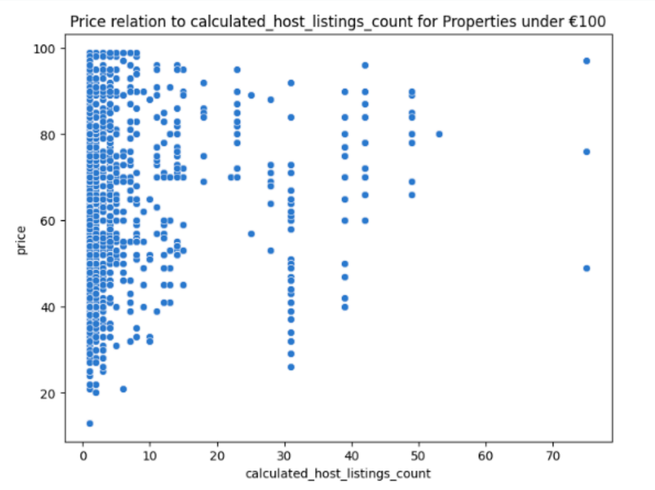
2.2 The Relationship between Price and Other Data

Some examples:

Structured data



Unstructured data



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3. Model Training

LINEAR

LINEAR

RANDOM FOREST

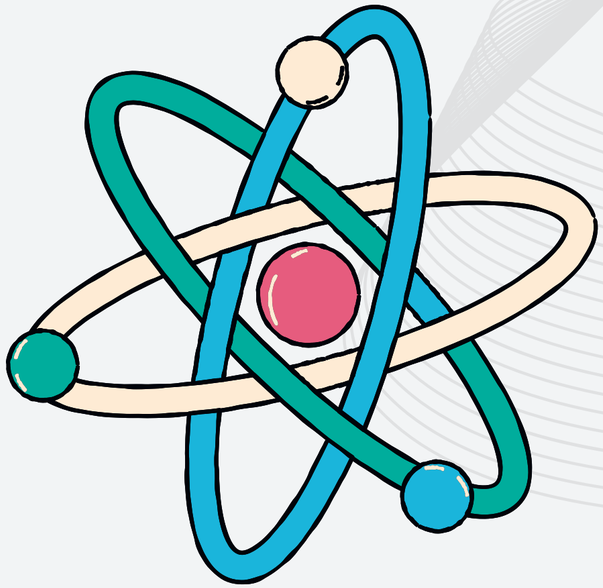
RAMDOM FOREST

LASSO

LASSO

RIDGE

RIDGE

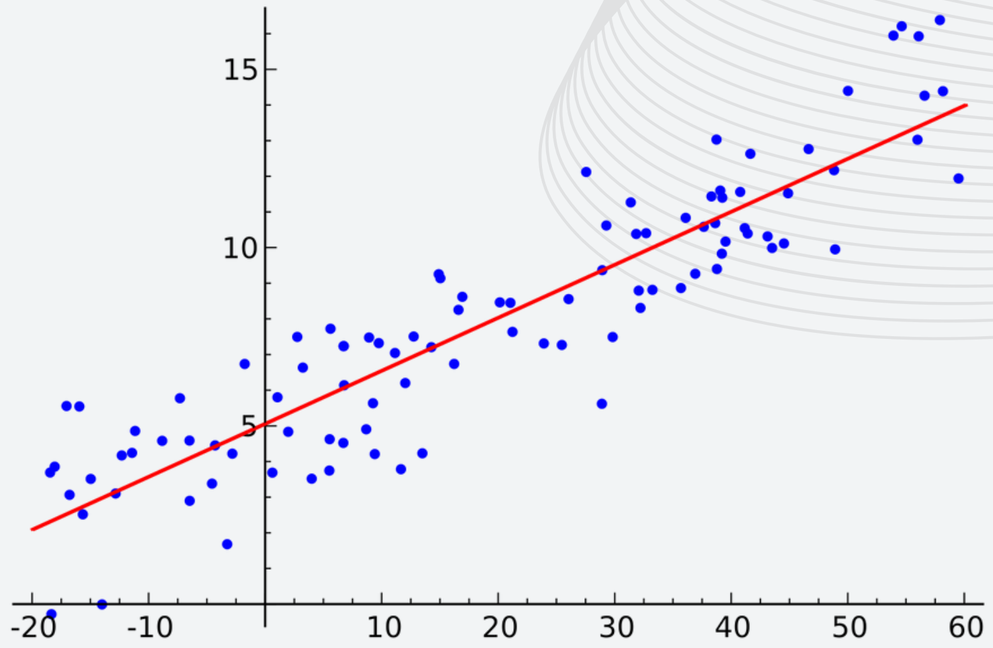


3.1 Linear Regression

Linear regression is a statistical method used to predict the linear relationship between a dependent variable and one or more independent variables.

- Features
  - Simple and easy to understand
  - High computational efficiency
  - Suitable for linear relationships

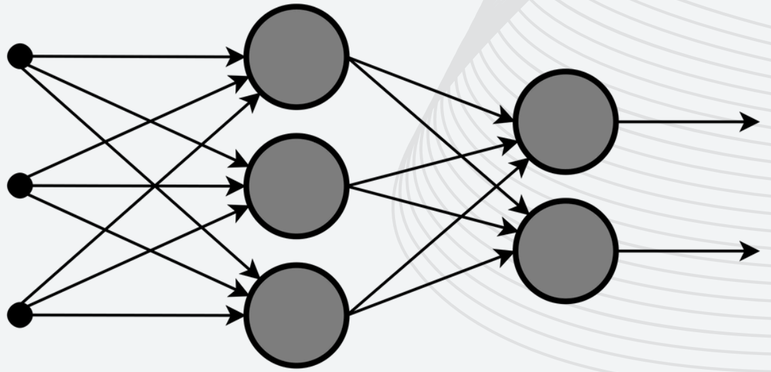
- Application Scenarios
  - Data has a linear relationship
  - Limited number of features
  - High need for interpretability



3.2 Random Forest Regression

Random Forest Regression is an ensemble learning method based on decision trees, combining multiple trees to improve prediction accuracy and stability.

- Features
  - Can handle non-linear relationships
  - High accuracy and stability
  - Robust to missing values and noise



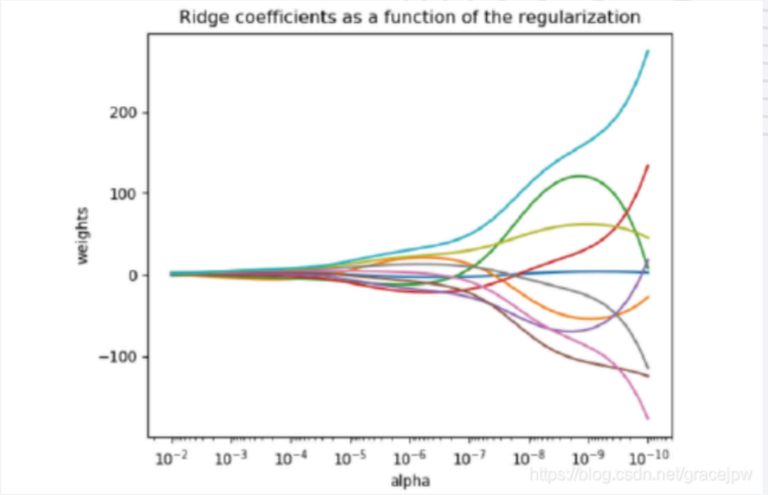
- Application Scenarios
  - Complex datasets with non-linear relationships
  - Large number of features
  - High demand for predictive accuracy

3.3 Ridge Regression

Ridge Regression is a linear regression with l2 regularization, penalizing the size of coefficients to reduce model complexity.

- Features
  - Can handle multicollinearity
  - Compresses all coefficients but does not make them zero
  - Improves the model's generalization ability

- Application Scenarios
  - Data with multicollinearity
  - Need to avoid overfitting
  - Necessity to retain all features

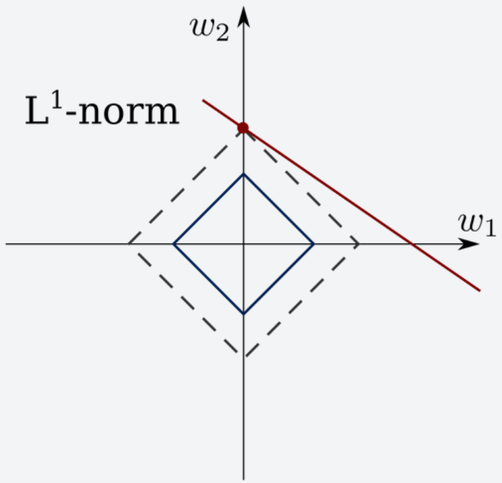


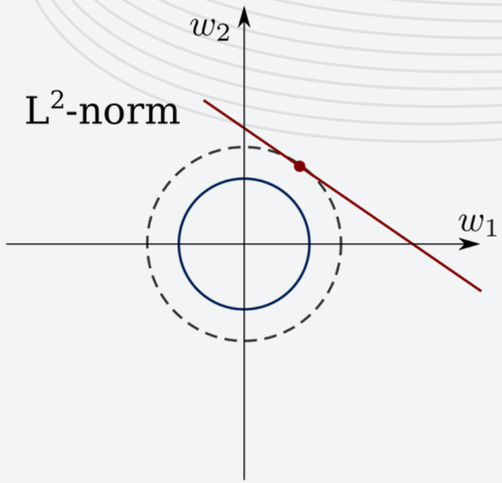
3.4 Lasso Regression

Lasso Regression (Least Absolute Shrinkage and Selection Operator) is a linear regression with l1 regularization that enables feature selection.

- Features
  - Can shrink the coefficients of irrelevant features to zero
  - Helps in feature selection
  - Reduces overfitting

- Application Scenarios
  - High-dimensional data
  - Need for feature selection
  - Presence of multicollinearity

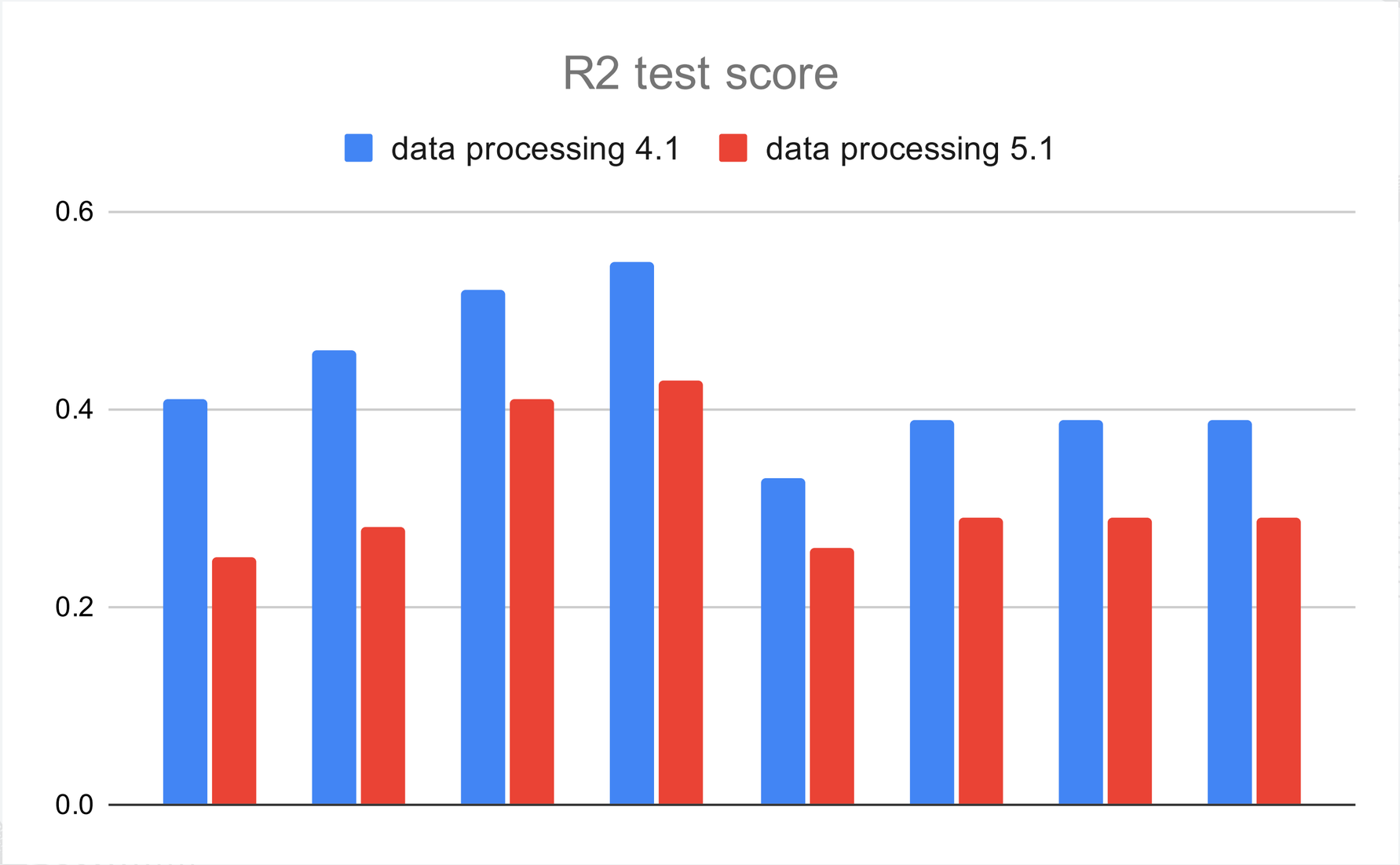




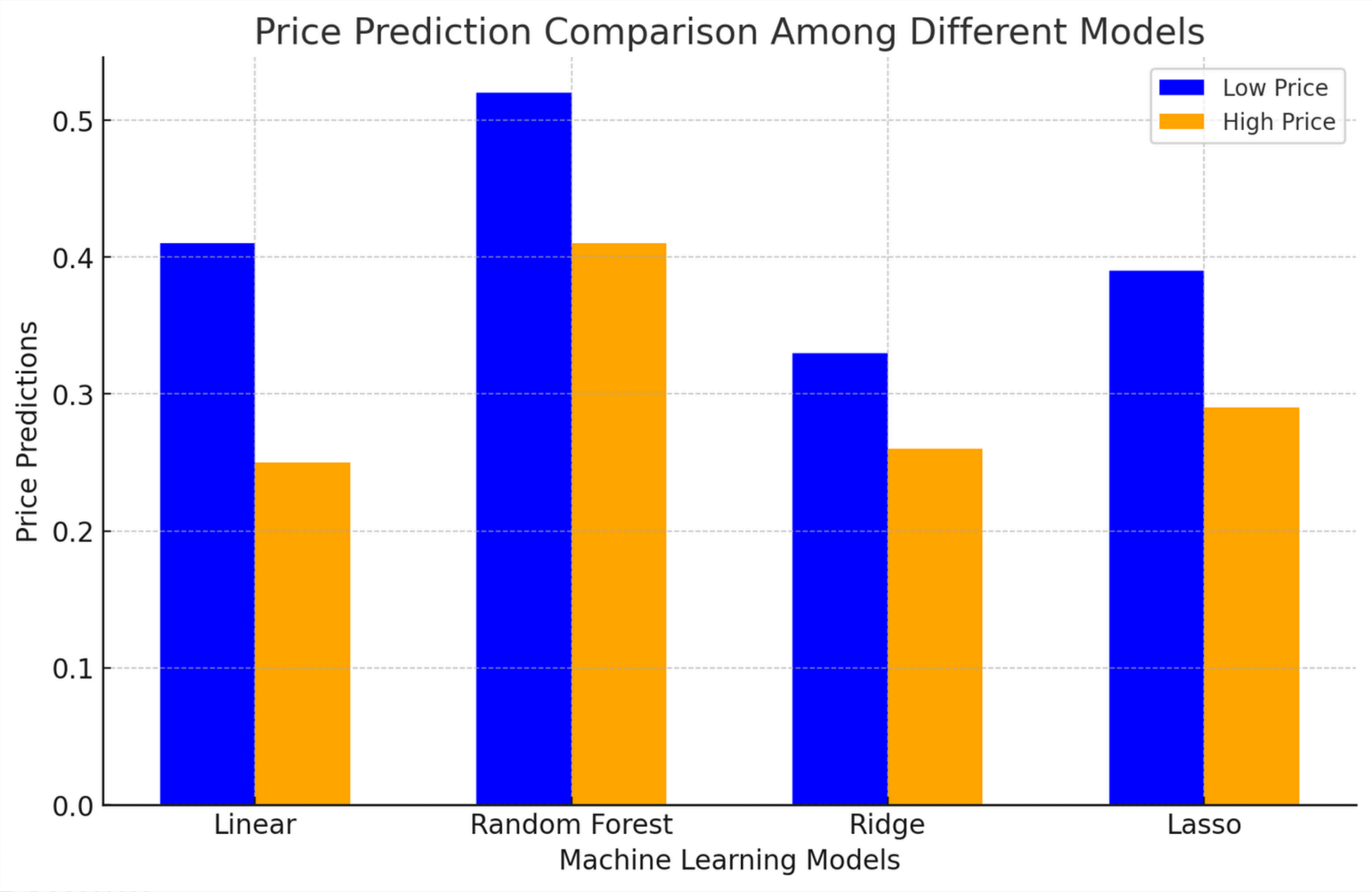
3.5 Improving Machine Learning Models

- KFold Cross-Validation
- Hyperparameter Tuning
  - GridSearchCV
- Random State
- Collinearity
  - principal component analysis (PCA)
- Categorical encoding

Performance



Performance



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Conclusion

- The properties have large differences in prices.
- Separating the dataset by price categories is useful for the analysis.
- The most interesting variables regarding price prediction are:
  - Neighborhood_group
  - Number of beds
  - Number of bedrooms
  - Number of baths
- If without them, in certain models R^2 score is even less than 0.1.
- The model Random forest regression have the best R^2 score for lower price, which is 0.52, and also the best R^2 score for higher price, which is 0.55.
- Using categorical ecnoded data did not improve the model.

Merci